

SUNJAE LEE

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🌐 <https://sunjae1294.github.io/> 🏠 <https://sites.google.com/view/fclab-skku>

RESEARCH & ENGINEERING INTERESTS

Building reliable computer-use agents and AI-native interfaces that enable AI to understand, operate, and reshape real-world software.

- **Computer-Use Agents**

GUI understanding and grounding, GUI world modeling, agent runtime and memory systems, planning, and long-horizon task execution.

Selected Work VeriSafeAgent [🔗](#), MobileGPT [🔗](#), AgentLens [🔗](#), MobiBench [🔗](#)

Ongoing GUI World Models, AI-Native Compiler and Framework, Self-Directed App Skill Discovery

- **AI-Native & Frontier Interfaces**

Agentic and generative interfaces, human-agent interaction, distributed and cross-device interfaces, and emerging device platforms.

Selected Work AgentLens [🔗](#), A-Mash [🔗](#), FLUID-XP [🔗](#), FLUID [🔗](#)

Ongoing Multimodal Educational Agents, AI-Native Compiler and Framework

- **Reliability & Evaluation**

Agent benchmarks and evaluation harnesses, action verification and guardrails for agents.

Selected Work VeriSafeAgent [🔗](#), MobiBench [🔗](#)

Ongoing Benchmarking Vibe Coding Agents, AI-Native Compiler and Framework

WORK EXPERIENCE

Assistant Professor, Sungkyunkwan University (SKKU) [🔗](#)

Department of Computer Science and Engineering

University Rankings: QS World University #108; THE Computer Science #78 (**3rd** in South Korea)

Rep. Korea

Mar. 2025 – Present

Founder & Solo Product Engineer, Marvis AI [🔗](#)

World-first production-grade smartphone-use agent for Android

Rep. Korea

2025 – Present

Co-Founder & Chief Technology Officer (CTO), Fluiz Corp. [🔗](#)

Distributed Android Operating Systems for Cross-Device Interaction Platform

Rep. Korea

July 2022 – Feb. 2025

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Ph.D. in Computer Science

September 2021 – Feb 2025

Daejeon, Rep.Korea

- Advisor: Prof. Insik Shin
- Thesis: Towards Ubiquitous and Intelligent Mobile Computing: From UI Distribution to Mobile AI Agents
- **Outstanding Ph.D. dissertation award** 🏆

Korea Advanced Institute of Science and Technology (KAIST)

M.S. in Computer Science

March 2019 – Feb 2021

Daejeon, Rep.Korea

- Advisor: Prof. Insik Shin
- Thesis: UI based Cross-device Screen distribution

Korea Advanced Institute of Science and Technology (KAIST)

B.S. in Computer Science

September 2012 – Feb 2019

Daejeon, Rep.Korea

SELECTED PUBLICATIONS

1. **MobiBench: Modular and Multi-Path Aware Benchmarking for Mobile GUI Agents** 
Youngmin Im, Byeongung Jo, Jaeyoung Wi, Seungwoo Beak, Sangeun Oh, Insik Shin, **Sunjae Lee**
Under Review
2. **AgentLens: Adaptive Visual Modalities for Human-Agent Interaction in Mobile GUI Agents** 
Jeonghyeon Kim, Byeongjun Joung, Junwon Lee, Joohyung Lee, Taehoon Min, **Sunjae Lee**
In Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology (UIST '26)
3. **VeriSafe Agent: Safeguarding Mobile GUI Agent via Logic-based Action Verification** 
Jungjae Lee, Dongjae Lee, Chihun Choi, Youngmin Im, Jaeyoung Wi, Kihong Heo, Sangeun Oh, **Sunjae Lee***, Insik Shin*
In Proceedings of the 31st ACM Int. Conf. on Mobile Computing and Networking (MobiCom '25)
4. **MobileGPT: Augmenting LLM with Human-like Memory for Mobile Task Automation** 
Sunjae Lee, Junyoung Choi, Jungjae Lee, Munim Hasan Wasi, Hojun Choi, Steven Y. Ko, Sangeun Oh, Insik Shin
In Proceedings of the 30th ACM Int. Conf. on Mobile Computing and Networking (MobiCom '24)
5. **FLUID-IoT: Flexible and Fine-Grained Access Control in Shared IoT Environments** 
Sunjae Lee*, Minwoo Jeong*, Daye Song, Junyoung Choi, Seoyun Son, Jean Y Song, Insik Shin
In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)
6. **Supporting Flexible and Transparent User Interface Distribution across Mobile Devices** 
Sangeun Oh, Ahyeon Kim, **Sunjae Lee**, Kilho Lee, Dae R. Jeong, Steven Y. Ko, and Insik Shin
IEEE Transactions on Mobile Computing
7. **MixMax: Leveraging Heterogeneous Batteries to Alleviate Low Battery Experience** 
Jaeheon Kwak, **Sunjae Lee**, Dae R Jeong, Arjun Kumar, Dongjae Shin, Ilju Kim, Donghwa Shing, Kilho Lee, Junkyu Lee, Insik Shin
In Proceedings of the 21st ACM int. Conf. on Mobile Systems, Applications, and Services (MobiSys '23)
8. **It is Okay to be Distracted: How Real-time Transcriptions Facilitate Online Meeting with Distraction** 
Seoyun Son, Junyoung Choi, **Sunjae Lee**, Jean Y Song, Insik Shin
In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)
9. **A-Mash: Providing Single-App Illusion for Multi-App Use through User-centric UI Mashup** 
Sunjae Lee, Hoyoung Kim, Sijung Kim, Sangwook Lee, Hyosu Kim, Jean Young Song, Steve. Ko, and Insik Shin
In Proceedings of the 28th ACM Int. Conf. on Mobile Computing and Networking (MobiCom '22)
10. **FLUID-XP: Flexible User Interface Distribution for Cross-Platform Experience** 
Sunjae Lee, Hayeon Lee, Hoyoung Kim, Sangmin Lee, Jeong Woon Choi, Yuseung Lee, Seono Lee, Ahyeon Kim, Jean Young Song, Sangeun Oh, Steven Y. Ko, and Insik Shin
In Proceedings of the 27th ACM Int. Conf. on Mobile Computing and Networking (MobiCom '21)
11. **FLUID: Flexible User Interface Distribution for Ubiquitous Multi-device Interaction** 
Sangeun Oh, Ahyeon Kim, **Sunjae Lee**, Kilho Lee, Dae R. Jeong, Steven Y. Ko, and Insik Shin
In Proceedings of the 25th ACM Int. Conf. on Mobile Computing and Networking (MobiCom '19, Best Paper 🏆)

ONGOING PROJECTS

GUI World Models for Computer-Use Agents

Ongoing

Developing world models that learn the dynamics of GUI state transitions and provide predictive representations for grounding, planning, and reliable interaction in computer-use agents.

Benchmarking End-to-End Vibe Coding Agents

Ongoing

Designing a realistic benchmark for end-to-end coding agents in modern AI-assisted development environments, bridging the gap between today's rigorously specified coding evaluations and real-world vibe coding.

AICampus: Multimodal Educational Agents for Live, Interactive Lectures

Ongoing

Building an AI-driven lecture platform where multimodal agents teach and interact with learners in real time through synchronized speech, live digital handwriting, slide generation, and adaptive instructional content.

AI-Native Compiler and Framework for Agentic Application Interaction

Ongoing

Developing an AI-native compiler and software framework that extends the software stack beyond human-computer interaction to agent-computer interaction.

Self-Directed Skill Discovery for Mobile GUI Agents

Ongoing

Developing methods for mobile GUI agents to autonomously explore applications, discover reusable task skills, and retrieve and compose those skills during planning and inference for new tasks.

PATENTS

1. **LLM-based noninvasive application execution system for mobile task automation, and method thereof** *Rep.Korea*
Application Number: 10-2744615 *Granted*
Insik Shin, Sunjae Lee *2024.12.16*
2. **User interface distribution method for multi-device interaction** *Rep.Korea*
Application Number: 10-2022-00091130 *Granted*
Insik Shin, Sangeun oh, Ahyeon Kim, Sunjae Lee *2022.02.28*
3. **Secure user interface distribution method for heterogeneous multi-device interaction** *Rep.Korea*
Application Number: 10-2019-0180075 *Published*
Insik Shin, Sangeun oh, Ahyeon Kim, Sunjae Lee *2019.12.31*
4. **Method of transmitting UI display information based on smart contract** *Rep.Korea*
Application Number: 10-2022-0026429 *Published*
Insik Shin, Daehee Lee, Sunjae Lee, Sangwook Lee *2022.02.28*
5. **Method of transeiving UI displays between user devices** *Rep.Korea*
Application Number: 10-2022-0026430 *Published*
Insik Shin, Daehee Lee, Sunjae Lee *2022.02.28*
6. **Method of transmitting UI display information to peripheral clients by a local server** *Rep.Korea*
Application Number: 10-2022-0026426 *Published*
Insik Shin, Daehee Lee, Sunjae Lee *2022.02.28*
7. **Secure user interface distribution method for heterogeneous multi-device interaction** *U.S*
Application Number: US17/852,146 *Published*
Insik Shin, Sangeun oh, Ahyeon Kim, Sunjae Lee *2022.11.17*

AWARDS

AI Champion Award (First Prize/630 teams) 🏆 <i>AI Champion competition hosted by Ministry of Science and ICT, South Korea</i>	Nov. 2025
Outstanding Ph.D. Dissertation Award, KAIST 🏆 <i>Thesis: Towards Ubiquitous and Intelligent Mobile Computing: From UI Distribution to AI Agent</i>	Feb. 2025
Minister's Award, Ministry of Science and ICT, South Korea 🏆 <i>Awarded during the 24th Mobile Technology Festival hosted by Korea Economic Daily</i>	Dec. 2024
CES 2023 Innovation Award 🏆 <i>Received as CTO of Fluiz Corp.</i>	Jan. 2023
Minister's Award, Ministry of Science and ICT, South Korea 🏆 <i>Awarded during the 22nd Mobile Technology Festival hosted by Korea Economic Daily</i>	Dec. 2022
MobiCom '19 Best Paper Award 🏆 <i>Awarded to Sangeun Oh, Ahyeon Kim, <u>Sunjae Lee</u>, Kilho Lee, Dae R. Jeong, Steven Y. Ko, Insik Shin</i>	Oct. 2019

SKILLS

Engineering: Full-Stack Engineering, Agent Systems Engineering, System Design

Languages & Platforms: Python, Java, C/C++, JavaScript; Android Open Source Project (AOSP), Linux

Communication: Korean (Native), English (Near-native), Human-AI Communication (Prompt & Context Engineering)